



Guardio Labs: Home Assignment

Introduction

This is a home assignment for a **Security Researcher** position at Guardio Labs.

The purpose of this home assignment is to allow the candidate to experience some of the tasks that Guardio security researchers perform and to allow common grounds for discussion in the next interview.

If you find that one of the questions is not strictly defined, you may use any assumptions you need, but make sure to explain why you made them.

Your submission should be a single doc file (format of your choosing) containing your full answers and any screenshot, code snippet you wish.

Include the task title and numbers when providing your answer.

If you get stuck, no worries! Feel free to reach out for hints (we have plenty).

Reversing And Detection

Attached is an CRX file for a Chrome extensions creating a dark theme for sites you visit, called **ProDark**.

Use any means you think relevant to analyse this extension and answer the following questions:

1. Apart from changing the background color, does this extension do other things? Explain what it does, how it is being done (in general pseudo-code).
2. Does this extension do bad stuff out of the box? What protection methods you see there to hide it's real intentions, and what is needed to activate that functionality (**bonus** - why this specific trigger is used here?)
3. What can this bad actor gain from these malicious activities? Think about possible ways to monetize.

4. Create a list of IOCs of any relevant type out of your research here (you can use any other resource relevant - just mention which if used)
5. Are there other extensions like this one? Created by the same actor? How would you suggest finding those?
6. Now that you know what this (and probably many other) extensions can do:
 - a. Suggest ways, techniques and methods to detect other malicious extensions that do the same bad things (assume you have access to the store and crx files of all extensions)
 - b. What other techniques you can think about that those bad actors can use to hide their real intentions? (Other than what was found here)

GOOD LUCK!